

TruePixel: Real vs AI Generated Image Classification

Abstract:

The rapid growth of generative AI, especially diffusion models, has made it easy to create highly realistic fake images that are difficult to distinguish from real ones. This raises concerns about misinformation and trust in digital media. In this project, we aim to build and evaluate a machine learning model that can automatically classify images as either real or AI-generated. The system is designed as a binary image classifier to detect synthetic content and support digital media authenticity.

The primary goal of this project is to develop a machine learning system that can classify AI-generated synthetic images. The project will focus on a comparative analysis of CNN based architectures such as EfficientNet-B0, ResNet-50 vs state-of-the-art Transformer based architectures.

First, we will implement and train **ResNet-50**[2] & **EfficientNet-B0**[3] models. Secondly, a **Vision Transformer Model** architecture will be implemented. These models will be trained on the same dataset and a benchmark analysis will be performed on different datasets.

To train and evaluate the models, we will use the **CIFAKE: Real and AI-Generated Synthetic Images** dataset from Kaggle [1], which contains 100,000 training images (50,000 real and 50,000 AI-generated) and 20,000 test images (10,000 per class). Later, we will also benchmark these models with generated data from popular AI Image Generators, for example DALL-E-3, Midjourney, etc. Moreover, a perturbation analysis will be carried out to test our models using techniques like JPEG compression, resizing, etc.

The project will be implemented in Python using the PyTorch deep learning framework. Data loading and preprocessing will be handled with Pandas and Pillow/OpenCV, while performance metrics such as FLOPs, training time, accuracy, precision, recall, and F1-score will be computed.

Finally, we will deploy the model as a lightweight web application that allows users to upload an image and instantly receive a classification indicating whether it is real or AI-generated.

References

[1] <https://www.kaggle.com/datasets/birdy654/cifake-real-and-ai-generated-synthetic-images/data>

[2] <https://docs.pytorch.org/vision/main/models/generated/torchvision.models.resnet50.html>

[3] https://docs.pytorch.org/vision/main/models/generated/torchvision.models.efficientnet_b0.html